



**Michigan  
Technological  
University**

## Course Syllabus

MA4535 – Nonlinear Dynamics and Chaos

College of Science and Arts, Fall 2026

Instructor: Allan, Struthers  
Office Location: 212 Fisher Hall  
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Schedule: <https://pages.mtu.edu/~struther/>  
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Office Hours: MF 11:00–11:50am and 1:00–1:50pm  
W 11:00–11:50am. I have dept meetings at 1pm.  
Just sstop by or check my schedule and make an appointment.

## Course Identification

Course Number: MA4535 -R01  
Course Name: Nonlinear Dynamics and Chaos  
Course Location: Fisher 126  
Class Times: M/W/F 12:00–12:50pm  
Prerequisites: ODE and Linear Algebra

## Course Description/Overview

Ordinary differential equations and dynamical systems via a modern geometric approach, including physical and engineering applications. May include chaotic phenomena and fractals.

## Course Resources

### Course Website(s)

- [Canvas](https://www.courses.mtu.edu) [www.courses.mtu.edu]
- GitHub <https://github.com/AllanStruthersMTU>

### Course Texts/Software

- Differential Equations for Scientists and Engineers by Struthers and Potter
  - Free from any university computer at <https://link.springer.com/book/10.1007/978-3-030-20506-5>
  - Available on Amazon <https://www.amazon.com/Differential-Equations-Scientists-Allan-Struthers/dp/3030205088>
- Mathematica, free to MTU students from <https://downloads.it.mtu.edu/>
- Julia, freeware available from <https://julialang.org/downloads/>

## **Course Learning Objectives**

Upon successful completion of this course, students should be able to

- Approximate and plot solutions of systems of ODEs.
- Compute and discuss the sensitivity of many systems of ODEs.
- Explain many topics from modern qualitative ODE theory.

## Grading Scheme

### Grading System

| <b>Grade</b> | <b>Percentage</b>                                                                                                                                                                                                                                                                                                                                                                           | <b>Grade points</b> | <b>Rating</b> |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|
| <b>A</b>     | 90 & above                                                                                                                                                                                                                                                                                                                                                                                  | 4.00                | Excellent     |
| <b>AB</b>    | 85 – 89                                                                                                                                                                                                                                                                                                                                                                                     | 3.50                | Very good     |
| <b>B</b>     | 80 – 85                                                                                                                                                                                                                                                                                                                                                                                     | 3.00                | Good          |
| <b>BC</b>    | 75 – 79                                                                                                                                                                                                                                                                                                                                                                                     | 2.50                | Above average |
| <b>C</b>     | 70 – 74                                                                                                                                                                                                                                                                                                                                                                                     | 2.00                | Average       |
| <b>CD</b>    | 65 – 69                                                                                                                                                                                                                                                                                                                                                                                     | 1.50                | Below average |
| <b>D</b>     | 60 - 64                                                                                                                                                                                                                                                                                                                                                                                     | 1.00                | Inferior      |
| <b>F</b>     | 59 and below                                                                                                                                                                                                                                                                                                                                                                                | 0.00                | Failure       |
| <b>I</b>     | Incomplete; given only when a student is unable to complete a segment of the course because of circumstances beyond the student's control.                                                                                                                                                                                                                                                  |                     |               |
| <b>X</b>     | Conditional, with no grade points per credit; given only when the student is at fault in failing to complete a minor segment of a course, but in the judgment of the instructor does not need to repeat the course. It must be made up by the close of the next semester or the grade becomes a failure (F). A (X) grade is included in the grade point average calculation as a (F) grade. |                     |               |

### Grading Policy

Grades will be based on the following:

| <b>Course Component</b>          | <b>Points</b> |
|----------------------------------|---------------|
| Homework                         | 45            |
| Projects/Exams                   | 45            |
| Attendance/Participation/Quizzes | 10            |
| <b>Total Points</b>              | <b>100</b>    |

### Late Assignments

Deadlines are enforced because solutions and code are discussed in class.

### Course Policies

- Attendance is important. Absences and late arrivals disrupt class.
- Ask questions in class. If something is unclear, others likely feel the same!
- Read material and do exercises before class.
- Maintain a respectful and professional learning environment.
- Collaboration and appropriate resource use on code/homework/presentations is encouraged. Not citing appropriately violates MTU Academic Integrity Policy 109-1

### AI Policy

Transparency in GenAI tool usage is mandatory, and instructors reserve the right to ask students to explain their work process. This can include prompts and AI-generated output.

GenAI changes fast. Course policies are provisional. If in doubt, consult the instructor. We want to learn how to use these tools appropriately and effectively.

### **Academic Integrity Rules**

Students should discuss homework, code, and projects. Students are individually responsible for individual work. Groups are jointly responsible for group work. Resources used, including but not limited to internet sites, AI tools, etc should be cited. Please do not distract the instructor and other students during class. Maintaining academic standards is important, academic misconduct may result in an appropriate conduct sanction/educational condition(s) imposed by the Office of Academic and Community Conduct and/or in a faculty-imposed academic penalty. For details, review the policy

[<http://www.admin.mtu.edu/usenate/policies/p109-1.htm>]

### **University Policies**

As part of the university's commitment to curricular quality, student work products may be used to evaluate how well Michigan Tech students are achieving programmatic learning outcomes. Findings are used to inform future course and curricular improvements to support student learning but are not used to evaluate specific students and individual instructors.

Michigan Tech has standard policies on academic misconduct and complies with all federal and state laws and regulations regarding discrimination, including the Americans with Disabilities Act of 1990. For more information about reasonable accommodations or equal access to education or services at Michigan Tech, please call the Dean of Students Office at 906-487-2212 or visit the [Student Disability Services website](#). More information is also available from the [Syllabi Policies webpage](#).

### **Course Schedule**

Student experience with prerequisites and interests will determine topics and sequencing. The final exam slot will be used for project presentations or an exam.

#### **Week 1**

*M      Syllabus review, introductions, and planning*  
*W      Introduction of Rosseler system and Sensitivity Computations*  
*F      Review of prerequisite ODE and Linear Algebra material*

#### **Week 2-14**

*MWF Sensitivity, periodicity, bifurcations, chaos, etc.*

#### **Finals Week**

The Registrar schedule final exam slot will be used for project presentations or a final exam.